1. Introduction to Android Architecture

Android is a Linux-based open-source operating system designed primarily for mobile devices like smartphones and tablets. Its architecture is layered and consists of:

- **Linux Kernel:** The foundation of Android, managing core system services like security, memory, process management, and hardware abstraction.
- **Hardware Abstraction Layer (HAL):** Interfaces between hardware and higher-level APIs, allowing Android to support a variety of hardware.
- **Native Libraries:** Libraries written in C/C++ for functionalities such as graphics (OpenGL), database (SQLite), and web rendering (WebKit).
- **Android Runtime (ART):** Runs Android applications using compiled bytecode. It replaces the older Dalvik VM for better performance and memory management.
- **Application Framework:** Provides APIs for app developers for managing UI, resources, notifications, telephony, location, etc.
- **Applications:** Built on top of the framework layer, including core apps like phone, SMS, browser, and user-installed apps.

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2. Android File Structure

Android uses a hierarchical file system, with important directories such as:

- **/system:** Contains core OS files and system apps.
- **/data:** Stores user data and installed app data.

- **/cache:** Temporary system files.
- **/sdcard or /storage:** External storage, accessible by user apps.
- **/proc, /dev, /sys:** Kernel and device files.

Each app gets its own sandboxed directory under /data/data/<package_name>/, isolated from others for security.

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3. Android Build Process

The build process converts source code into an installable APK package:

- **Source Code:** Java/Kotlin code + resources (XML, images).
- **Compilation:** Java/Kotlin code is compiled into bytecode (.class files).
- **Dexing:** Bytecode converted to Dalvik Executable (.dex) format for ART.
- **Packaging:** Resources and .dex files are bundled into an APK.
- **Signing:** APK is digitally signed with a developer’s key.
- **Optimization:** Optional steps like ProGuard for code shrinking and obfuscation.

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4. Android App Fundamentals

Android apps consist of four main components:

- **Activities:** UI screens, user interactions.
- **Services:** Background processes without UI.
- **Broadcast Receivers:** Listen for system-wide events.
- **Content Providers:** Manage app data and share it with others.

Apps run in their own sandboxed environment with unique user IDs for security and isolation.

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5. Android Security Model

Key aspects of Android security:

- **Sandboxing:** Each app runs in a separate Linux process with a unique user ID.
- **Permissions:** Apps must declare permissions to access sensitive resources (camera, location).
- **App Signing:** Ensures app authenticity and integrity.
- **SELinux:** Enforces mandatory access controls on system resources.

- **Verified Boot:** Ensures device boots only with trusted software.
- **Google Play Protect:** Scans apps for malware.

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6. Device Rooting

- **Rooting** is the process of gaining superuser (root) access to an Android device, bypassing built-in security restrictions.
- Root access allows modifying system files, installing custom ROMs, and accessing features blocked by the manufacturer.
- **Risks:** Voids warranty, increases vulnerability to malware, and can cause instability.
- Rooting is often done by exploiting vulnerabilities or flashing modified system images.